3.a

→ $P \in (0,1)$

→ Fraction P of examples in validation set are positive (with 1 as label)

→ Fraction $1-P$ of examples in validation set are negative (0 label)

$$\text{Accuracy} \triangleq \frac{\#\text{ Examples that are predicted correctly by classifier}}{\#\text{ Examples}}$$

i) For any dataset with p fraction of positive examples and $1-P$ fraction of negative examples, there exists a trivial classifier with accuracy $(1-P)$.

Imagine a trivial classifier C_T which predicts every input as negative (ie 0 label) irrespective of what the input is, in any given dataset.

Since by definition, any data set we are considering has $1-P$ fraction of negative examples/inputs, our trivial classifier C_T classifies all of those $(1-P)$ fraction inputs correctly as negative. Hence it will have the accuracy of $(1-P)$. Therefore any data set with P fraction of positive examples and $(1-P)$ negative examples, there exists a trivial classifier of accuracy $(1-P)$.

If 'n' is # of inputs in a dataset with P positive and $1-P$ negative as fractions,

$$\text{Accuracy} = \frac{n(1-P)}{n} = \underline{1-P}$$

3. a

$$A_1 \triangleq \frac{TP}{TP+FN} \qquad A_0 \triangleq \frac{TN}{TN+FP}$$

Balanced Accuracy $= \bar{A} \triangleq \frac{1}{2}(A_0 + A_1)$ — ①

$$A = \frac{TP+TN}{TP+FN+TN+FP}$$

ii) Show $p = \dfrac{TP+FN}{TP+TN+FP+FN}$

* p is the fraction indicating the real positive samples irrespective of our classifier's prediction.
* out of True positive (TP), True negative (TN), False positive (FP) and False negative (FN), the fraction denoting all positive samples/examples is $\boxed{TP + FN}$

$FN \rightarrow$ positive classified as negative

$TP \rightarrow$ positive classified as positive

Let 'n' be total no. of samples, the fraction of positive samples in these 'n' is nothing but

$$p = \frac{\text{total no. of positive samples}}{\text{total no. of all the samples.}}$$

$$p = \frac{(TP+FN)\times \cancel{n}}{(TP+TN+FP+FN)\times\cancel{n}}$$

$$\boxed{\text{therefore } p = \frac{TP+FN}{TP+TN+FP+FN}}$$

3.00

ii) contd....

Show $A = p.A_1 + (1-p) A_0$.

$p.A_1 + (1-p) A_0$

$$= \frac{(TP+FN)}{(TP+TN+FP+FN)} \cdot \frac{TP}{(TP+FN)} + \left(1 - \frac{(TP+FN)}{TP+TN+FP+FN}\right) \times \frac{TN}{TN+FP}$$

$$= \frac{(TP+FN)}{(TP+TN+FP+FN)} \cdot \frac{TP}{(TP+FN)} + \left(\frac{\cancel{TP}+TN+FP+\cancel{FN}-\cancel{TP}-\cancel{FN}}{TP+TN+FP+FN}\right) \times \frac{TN}{TN+FP}$$

$$= \frac{(TP+FN)}{TP+TN+FP+FN} \cdot \frac{TP}{TP+FN} + \frac{\cancel{TN+FP}}{TP+TN+FP+FN} \cdot \frac{TN}{\cancel{TN+FP}}$$

$$= \frac{(TP+FN)}{TP+TN+FP+FN} \cdot \frac{TP}{TP+FN} + \frac{TN}{TP+TN+FP+FN}$$

$$= \frac{(TP)^2 + FN.TP}{(TP+TN+FP+FN).(TP+FN)} + \frac{TN}{TP+TN+FP+FN}$$

$$= \frac{TP^2 + FN.TP + TN.TP + FN.TN}{(TP+TN+FP+FN).(TP+FN)}$$

$$= \frac{TP(TP+FN) + TN(TP+PN)}{(TP+TN+FP+FN)(TP+FN)}$$

3. a. ii Contd...

$$= \frac{(TP \cancel{+FN})(TP+TN)}{(TP+TN+FP+FN)(TP\cancel{+FN})}$$

$$= \frac{TP+TN}{TP+TN+FP+FN}$$

$$= A.$$

Therefore our starting point $p \cdot A_1 + (1-p)A_0 = A$

~~proved above~~ has been proved.

3.a

iii Show trivial classifier in (i) has a balanced accuracy of 50%

Definition of balanced accuracy is given as $\bar{A} \triangleq \frac{1}{2}(A_0 + A_1)$.

Lets plug-in $A_0 = \frac{TN}{TN+FP}$, $A_1 = \frac{TP}{TP+FN}$

$$\bar{A} \triangleq \frac{1}{2}(A_0 + A_1)$$

$$\triangleq \frac{1}{2}\left[\frac{TN}{TN+FP} + \frac{TP}{TP+FN}\right]$$

our trivial classifier ~~only~~ classified every input as negative, it can't classify any input as positive.

Hence True positive $\to TP = 0$ and

False positive $\to FP = 0$.

plugging $TP = 0$, $FP = 0$ in above $\bar{A}$

$$\bar{A} \triangleq \frac{1}{2}\left[\frac{TN}{TN} + \frac{0}{FN}\right]$$

$$\bar{A} \triangleq \frac{1}{2}[1+0] = \frac{1}{2} = 50\%$$

Hence balanced accuracy of our trivial classifier in (i) is 50%.